

ePC-SAFT For Produced-Water Lithium Extraction

Rezaee DES/TOPO Li/Na bridge for thermodynamic and process-systems integration

Chemical engineering technical review

2026-05-08

0. Technical Thesis

This deck makes one chemical-engineering argument:

Produced-water lithium extraction needs a thermodynamic model that maps basin chemistry into solvent-extraction transfer variables. The active Li/Na bridge is Rezaee DES/TOPO after divalent pretreatment; ePC-SAFT is evaluated as the equilibrium and diagnostics backbone for PrOMMiS and IDAES.

- Feed basis: southern Arkansas Smackover brine.
- Solvent chemistry: Rezaee TBAC/decanoic-acid DES with TOPO coextractant.
- Thermodynamic target: phase-specific reactive equilibrium, electrolyte diagnostics, and ion-specific distribution.
- Process target: staged extraction, surrogate generation, and IDAES costing.

Restrict quantitative claims to source-backed literature, generated validation tables, and explicitly labeled model outputs. Divalent ions are feed-context species only; the active objective is Li extraction from Na after divalent pretreatment.

Open with the implementation need, not with a generic lithium motivation. The audience should understand within the first minute that this is a case for thermodynamic integration, not a broad solvent-discovery survey.

1. Basin Thermodynamic Basis

1.1 Produced Water Is Not One Feed

Basin	Representative lithium signal	Representative TDS signal	Why it matters
Appalachian	Marcellus subsets reach 127–205 mg/L Li	Often >100,000 mg/L TDS	High Li, chemically variable
Permian	Wolfcamp average near 14 ppm Li	<25–140 g/L TDS range	High water volume, lower Li grade
Williston	Brines can exceed 100 mg/L Li	Bakken/Three Forks near 308 g/L TDS	Salinity can dominate pretreatment
Smackover	Observed Li about 1–477 mg/L	156,000–340,000 mg/L TDS	Premium but thermodynamically hard

The case study starts with basin screening because the same extraction train will not behave the same across these feeds.

This is the first premise. If produced water were a single feed, a simple recovery factor might be defensible. It is not a single feed, so the model needs to carry chemistry.

1.2 Why Smackover Is The Flagship Case

Why this location makes sense

- Southern Arkansas Smackover combines high Li with very high salinity, so the chemistry is both valuable and hard.
- Smackover and Marcellus are the more favorable U.S. plays on major-metals-to-lithium and divalent-to-lithium ratios in the cited screening work.
- Arkansas already has commercial bromine and brine-handling infrastructure.
- That combination connects chemistry, location, and process deployment.

Signal	Smackover basis
Li range	observed wells span about 1–477 mg/L
TDS range	observed samples span 156,000–340,000 mg/L
Median TDS	current observation-set median is 305,000 mg/L
Infrastructure	existing Arkansas bromine/brine operations

The key line is: Smackover wins not because it is easy, but because it is valuable and thermodynamically unforgiving. That is exactly where an electrolyte EoS earns its keep.

1.3 Smackover Feed Basis

Case	Row	Li (mg/L)	TDS (mg/L)	Na/Ca/Mg (mg/L)	Role
Low observed	SH-2	11.7	156,000	37,100 / 13,900 / 2,250	Lower source bound
Base observed	MS-2	168	305,000	64,100 / 36,900 / 3,310	Median-TDS base proxy
High observed	BR-2	252	340,000	70,800 / 39,900 / 3,130	High-TDS/high-Li proxy

These are source rows from the USGS southern-Arkansas Smackover data release after duplicate and blank rows are excluded. The separate 100/200/400 mg/L Li rows are sensitivity cases, not direct well samples.

This slide shows true source composition without implying that the Rezaee extraction experiments were performed on this Smackover sample.

1.4 Selected Base Brine

Field	Value
Source row	MS-2 / MSPU 4-W1
Formation	Smackover
Li	168 mg/L
TDS	305,000 mg/L
Na / Ca / Mg	64,100 / 36,900 / 3,310 mg/L
Cl / Br	174,000 / 2,700 mg/L

The feed is Smackover-specific. The extraction chemistry is Rezaee DES/TOPO. The model couples a source-backed brine to a literature-backed Li/Na extraction chemistry without claiming a fully calibrated Arkansas extraction experiment.

This slide makes the case concrete while preserving the source-vs-chemistry boundary.

1.5 Critical Minerals And REE Boundary

Group	What the Smackover source table supports	How to present it
Reported trace fields	Fe, Mn, Zn, Al, As, Be, Cd, Co, Cu, Cr, Hg, Mo, Ni, Pb, U, V	Use as trace/pretreatment context, with censoring caveats
REE	not_reported in the source table	Do not infer; make this a future-data need

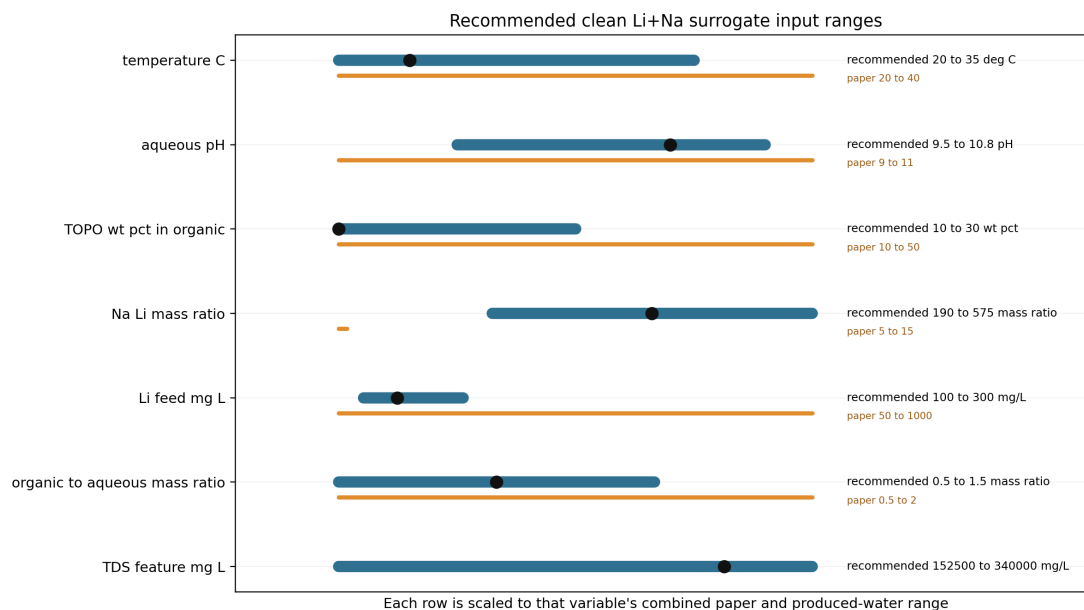
Many trace-metal values are censored thresholds, so avoid treating medians as exact resource grades. The key message is disciplined: REE are not source-supported for this feed yet.

1.6 Variables The Case Must Vary

Axis	Why it matters	Process consequence
Li concentration	Sets value and concentration driving force	Changes recovery target and stage count
Na/Li ratio	Defines the clean post-pretreatment competition problem	Changes Li/Na selectivity and sodium co-transfer
pH and temperature	Rezaee reports these as major response drivers	Moves extraction and selectivity at the chemistry level
TOPO loading	Controls ligand capacity and solvent cost	Changes (D _i), selectivity, and solvent inventory
O/A ratio	Directly affects stage transfer	Becomes a design and optimization variable
TDS / salinity feature	Carries produced-water severity into the process model	Sets surrogate validity and pretreatment/costing context

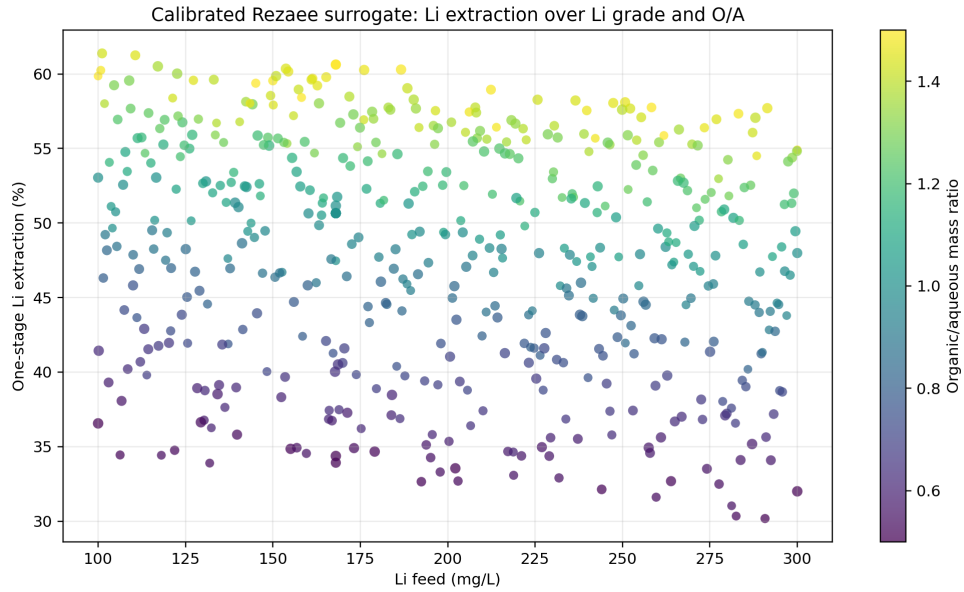
The raw Smackover brine is still the source site. The extraction model starts after pretreatment, so Mg, Ca, Sr, and Ba are not active extraction variables in this Li/Na bridge.

1.7 Clean Li/Na Surrogate Input Space



This is the input contract for the next surrogate run. It separates Rezaee's paper-tested ranges from the produced-water extension needed for the selected Smackover clean Li+Na stream. Do not present these bounds as completed response surfaces until the concrete run script generates the extraction outputs.

1.8 Surrogate Readiness Surface



This scaffold response surface is for process-integration readiness. It checks that the TDS, Li, and O/A design can be carried through transfer variables, staged recovery, and costing before the calibrated Rezaee/ePC-SAFT surrogate rows replace it.

2. ePC-SAFT Thermodynamic Role

2.1 The Model Gap

What weak placeholders can do

- Reproduce a narrow demonstration point
- Provide a quick stage recovery factor
- Run fast inside a flowsheet
- Help debug material balances

What they cannot defend

- Basin-to-basin extrapolation
- Hypersaline electrolyte behavior
- Solvent and O/A sensitivity
- Ion-specific distribution behavior

- Trust-region bounds for optimization

The process model needs a physical equilibrium layer before the surrogate or optimizer is meaningful.

Be careful not to oversell. This is a replacement for unsupported transfer assumptions, not a claim that ePC-SAFT solves kinetics, hydraulics, or plant economics by itself.

2.2 What ePC-SAFT Adds

Modeling need	ePC-SAFT contribution	Why it is novel for this case
Hypersaline brine	Electrolyte activity and phase-equilibrium structure	Salinity becomes a modeled variable, not a correction factor
Aqueous/organic split	Phase compositions and phase fractions	The extraction stage gets a thermodynamic target
Ion partitioning	Distribution ratios and selectivity metrics	Li movement can be separated from bulk salt movement
Sensitivity surfaces	Response to TDS, O/A, solvent, and feed composition	Surrogates can be trained inside a meaningful region
Diagnostics	Stability, convergence, and phase-distance checks	Failed or collapsed points do not silently become process data

The novelty claim is reusable insight. The model lets the project ask where the chemistry should work, why the answer changes across basins, and what variables the process model should receive. For this deck, those variables are transfer variables, not equipment performance coefficients.

2.3 Rezaee 2026 As A Parameter-Fitting Smoke Test

What Rezaee gives	How this deck should use it
2025 extraction responses and SI phase-composition rows	Source-backed Li/Na bridge targets
2026 DES/TOPO/RLi/RNa PC-SAFT parameters and organic (k_{ij}) values	Package-facing parameter payload

What Rezaee gives	How this deck should use it
2026 reaction constants and phase-specific ion-exchange equations	Reactive-equilibrium diagnostic target
Aqueous electrolyte phase modeled with ePC-SAFT	Activity and stability diagnostics for the brine side

Rezaee is now the active Li/Na bridge. Direct published-constant closure is still a documented activity-reference gap.

This slide prevents overclaiming. The source-regressed bridge is presentation-facing; the direct published-constant thermodynamic replay remains diagnostic until the activity-reference gap is closed.

2.4 Solvent Candidate Scorecard

Priority	System	Role	Status
1	TBAC + decanoic-acid DES + TOPO	Flagship Li/Na bridge	Rezaee source targets, parameter payloads, bridge outputs, and ePC-SAFT diagnostics exist
2	HBTA + TOPO + sulfonated kerosene	Conventional comparison	Strong field-water evidence, but missing full organic parameter payload
3	D2EHDTPA + BuPhen + octanol + n-dodecane	Best non-HBTA backup	Strong multication selectivity, blocked by missing parameter set
4	TBP + FeCl ₃ /HCl	Mechanistic backup	Useful high-Mg comparison, process-burden and parameter gaps

This ranking keeps the active benchmark source-backed and package-facing while still identifying conventional comparison solvent families.

2.5 Limitation Of Fixed-Coefficient Crossflow

Placeholder limitation

- The legacy Jang-style crossflow placeholder reaches only 39.96% cumulative Li extraction after ten contacts.
- Sodium still reaches 38.05% cumulative extraction in that same chain.
- It is a limitation baseline, not a flagship produced-water case.

Why it was not enough

- fixed surrogate chemistry
- weak Li-over-Na separation
- no bridge to trust-region sampling or ALAMO

This slide creates the contrast. The older result is not embarrassing; it is the baseline that proves why a better thermodynamic/chemistry architecture is needed.

3. Selective Extraction Model

3.1 One-Stage Selective Split

MS-2 O/A = 1 stage result

- Li extraction: 45.3660%
- Na extraction: 8.9780%
- ($D_{\text{Li}} = 0.8304$)
- ($D_{\text{Na}} = 0.09864$)
- ($S_{\text{Li/Na}} = 8.4185$)
- Trust label: `anchored_by_rezaee_real_brine_high_na_li`

Source-regressed Rezaee Li/Na bridge after divalent pretreatment; not a divalent reactive LLE flash.

What changed from the old wrapper

- The active bridge now uses Rezaee DES/TOPO source responses instead of the old HBTA/TOPO placeholder.
- The package evaluates ePC-SAFT electrolyte stability and PC-SAFT/ePC-SAFT activity diagnostics.
- The old HBTA/TOPO wrapper remains only as comparison and future parameterization context.

- The current stage-output table carries extraction, distribution, selectivity, and validity labels.

The important contrast is selective transfer after divalent pretreatment: lithium moves more strongly than sodium, and the bridge remains explicitly source-regressed until direct Rezaee thermodynamic closure is resolved.

3.2 Rezaee Bridge Closure

Evidence item	Source artifact	Value	Use
Extraction target rows	<code>rezaee_2025_extraction_target_summary.csv</code>	0.39	Source-regressed bridge fit
SI phase-composition rows	<code>rezaee_2025_extraction_equilibrium_summary.csv</code>	0.26	Reactive-equilibrium diagnostic
Median source (D_{Li})	<code>rezaee_li_na_distribution_coefficients.csv</code>	0.6017	Surrogate prior
Median source (D_{Na})	<code>rezaee_li_na_distribution_coefficients.csv</code>	0.1925	Surrogate prior
Bridge Li logit RMSE	<code>rezaee_li_na_distribution_bridge_report.csv</code>	0.154	Fit diagnostic
Bridge Na logit RMSE	<code>rezaee_li_na_distribution_bridge_report.csv</code>	0.1615	Fit diagnostic

This is the current presentation-facing closure: Li/Na staged transfer from a Rezaee source-regressed bridge. Ca, Mg, and other divalent ions remain pretreatment context, not active extraction targets.

3.3 Multi-Stage Performance

Stage metric	Rezaee bridge value
Stage 1 Li extraction	45.3660%
Stage 1 Na extraction	8.9780%
Stage 3 Li recovery	83.6925%
Stage 3 Na recovery	24.5883%

Three-stage bridge interpretation

- The stage chain is source-regressed from Rezaee Li/Na responses.
- The Smackover row is a feed-context projection, not a measured Arkansas extraction experiment.
- The result is strong enough for PrOMMiS/IDAES handoff, not investment-grade TEA.

The three-stage value is useful as a design signal and bridge output. It should not be presented as direct published-constant ePC-SAFT LLE closure.

3.4 Thermodynamic Insight

The old question

Can this solvent extract lithium from this brine?

The stronger question

Where should this chemistry work, how does the answer change with basin chemistry, and what stage variables should enter a process model?

Why ePC-SAFT changes the question

- It turns feed chemistry into equilibrium states.
- It lets salt load and O/A ratio become computable sensitivities.
- It provides distribution and selectivity variables, not only recovery percentages.

This is the main “novel insight” slide. The idea is not just a better lithium number. It is a more reusable way to reason from location to chemistry to process design.

4. PrOMMiS / IDAES Process Bridge

4.1 From Equilibrium To Transfer Variables

Process-model need	ePC-SAFT-derived quantity	Use in PrOMMiS / IDAES
Feed state	(z _i), TDS, Li/Na/Mg/Ca/Sr/Ba/Cl basis, temperature	Basin-specific solvent-extraction inlet
Phase split	aqueous and organic compositions, phase fraction	Equilibrium target or closure for a stage
Transfer strength	(D _i = C _{i,org} /C _{i,aq})	Distribution behavior in material balances
Selectivity	(S _{Li/Na})	Separation constraint for the active Li/Na showcase

PrOMMiS and IDAES do not just need a lithium recovery number. They need a defensible map from produced-water chemistry to stage transfer variables.

This is the concrete software interface. Percent extraction is the audience-friendly number; the unit model needs compositions, phase splits, distribution ratios, selectivity, and diagnostics.

4.2 Transfer-Variable Contract

For process integration, the ePC-SAFT output should be treated as the following structured contract:

Variable	Definition	Why PrOMMiS / IDAES needs it
($_i$)	stage removal ($= (C_{\{i,in\}} - C_{\{i,out\}}) / C_{\{i,in\}}$)	Links chemistry to process-wide recovery objectives
(D_i)	$(C_{\{i,org\}} / C_{\{i,aq\}})$	Controls staged transfer in material balance equations
($S_{\{Li/Na\}}$)	Selectivity ratio from stage/phase outputs	Enables solvent and design comparison without refitting fixed coefficients
Validity flags	residuals, phase flags, domain bounds	Prevents brittle points from entering surrogate or optimizer datasets

This keeps the story concrete: ePC-SAFT computes the transfer map; PrOMMiS/IDAES consumes it.

This slide is the strongest technical anchor for implementation reviewers because it is explicit about what gets passed across process boundaries.

4.3 Why This Belongs In PrOMMiS / IDAES

- 1. Offline engine** Run ePC-SAFT outside the flowsheet to generate equilibrium tables over feed chemistry, O/A ratio, and solvent conditions.
- 2. Surrogate layer** Fit ALAMO or another surrogate inside a documented trust region so optimization is fast and bounded.
- 3. External function** Expose high-fidelity calls for studies where direct thermodynamic solves are worth the cost.

4. Process model Use PrOMMiS/IDAES for staged extraction, recycle, stripping, costing hooks, and optimization.

This is the recommended maturity ladder. Do not pitch direct coupling as the only path. The lowest-risk first step is an offline data generator that feeds a bounded surrogate.

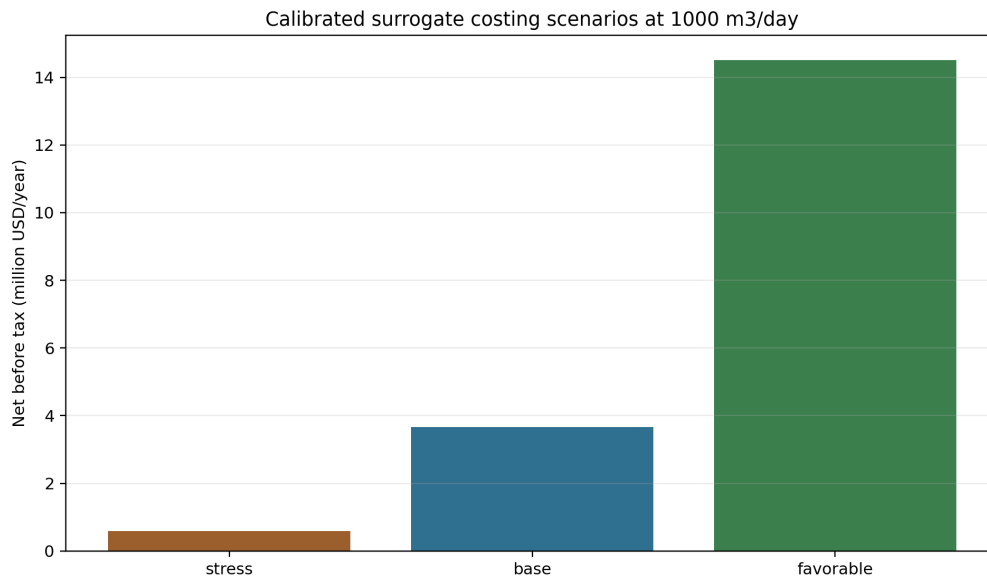
4.4 Formal IDAES Costing Scaffold

Costing block	Current basis	Needed before formal TEA
Feed and annual basis	Smackover base feed; 100/1000/10000 m3/day scenarios	approved flowrate and uptime
Pretreatment	IDAES/Pyomo scenario expression from model inputs	removal target, reagent use, waste handling
Extraction/contacting	three-stage Rezaee DES/TOPO Li/Na bridge	contactor sizing, residence time, solvent inventory
Solvent makeup/loss	low/base/high placeholder per m3 feed	DES/TOPO prices and solvent-loss rate
Concentration and Li2CO3 precipitation	bridge recovery 83.6925%	evaporation duty, carbonate reagent, product spec

The formal IDAES/Pyomo layer computes production, revenue, annual operating cost, annualized contactor CAPEX, breakeven Li2CO3 price, and net-before-tax scenario values.

The assumptions are scenario placeholders, not vendor quotes.

4.5 MSContactor/Costing Readiness Check



The Rezaee clean Li/Na transfer table now drives a PrOMMiS-side IDAES/Pyomo costing block with stress, base, and favorable cases. This verifies the handoff shape and cost-accounting hooks; the calibrated surrogate will replace the scaffold response values.

4.6 Implementation Requirements

Add ePC-SAFT as the thermodynamic backbone

- property/equilibrium service for electrolyte LLE
- documented species and feed basis
- diagnostics carried with every result
- reproducible dataset generation

Use PrOMMiS/IDAES for the process system

- solvent-extraction stage balances
- multi-stage contactors
- surrogate and external-function interfaces
- flowsheet optimization and costing hooks

The sell is simple: ePC-SAFT supplies the equilibrium evidence; PrOMMiS/IDAES turns that evidence into a process model.

This slide is the ask. It should leave no ambiguity about what you want: implement ePC-SAFT into the ecosystem so these cases can move from isolated scripts into reusable process models.

4.7 Remaining Engineering Work

Priority	Work item	Why it matters
1	Resolve the Rezaee activity-reference gap	Determines whether direct published-constant reactive equilibrium can replace the bridge
2	Replace scaffold responses with calibrated Rezaee/ePC-SAFT surrogate rows	Converts the clean Li+Na input matrix into accepted extraction, (D _i), selectivity, and validity rows
3	Upgrade the Rezaee-named PrOMMiS MSContactor case from algebraic transfer to unit-model validation	Tests convergence and material-balance consistency under staged operation
4	Replace Class-5 costing assumptions with engineering ranges	Improves economic credibility before TEA-style interpretation
5	Keep HBTA/TOPO parameterization as a comparison lane	Preserves the conventional field-water chemistry path without blocking Phase 0-9

This gives you a transparent ending if someone asks what is already proven versus what remains. It also turns the next work into a clear roadmap rather than a vague promise.

5. Backup: Sensitivity Expansion

5.1 Next-Step Sensitivity Expansion

- Re-run the Rezaee bridge across basin-style feed envelopes:
 - Appalachian high-Li case
 - Permian volume-driven low-grade case
 - Williston high-TDS case
 - Smackover premium-brine case

- Keep the next design-of-experiments table in Markdown so trust-region bounds and surrogate regions are explicit.
- Start from the scaffold response table and replace the response columns with calibrated model outputs.
- Add a PrOMMiS/IDAES interface table for every generated variable: units, species basis, source basis, and validity limits.
- Keep the narrative in one place: screening, thermodynamics, selective result, and process-design interface.

This backup slide is the roadmap for the next systematic screening campaign. It should be shown when someone asks how this becomes more than one case.